Topic 3: Spectrum Representation

Summary based on J. H. McClellan, R. W. Schafer, and M. A. Yoder, Signal Processing First, Prentice Hall, 2003.

Topics: Spectrum representation, sums of sinusoids, beat notes, amplitude modulation, periodic waveforms, Fourier series, Fourier analysis and synthesis, spectrograms, and chirp signals.

> *Key idea. A signal's spectrum identifies its sinusoidal components and their complex amplitudes. Fourier analysis decomposes periodic signals into harmonically related components, while time-frequency representations show how spectral content evolves over time.*

3.1 The spectrum of a sum of sinusoids

Sinusoids are the basic building blocks used to construct more complex signals. The most general and powerful way to produce new signals from sinusoids is linear addition:

$$x(t) = \sum_{k=1}^{N} A_k \cos(\omega_k t + \phi_k),$$

where each amplitude, phase, and frequency can be chosen independently. The signal can also be represented using the phasor of each sinusoidal component:

$$x(t) = \Re\left\{\sum_{k=1}^{N} X_k\, e^{j\omega_k t}\right\}, \qquad X_k = A_k e^{j\phi_k}.$$

Each phasor represents the amplitude and phase of a rotating phasor at frequency ω_k. Using the inverse Euler formula,

$$A_k \cos(\omega_k t + \phi_k) = \frac{1}{2} X_k e^{j\omega_k t} + \frac{1}{2} X_k^* e^{-j\omega_k t}.$$

Therefore,

$$x(t) = \sum_{k=1}^{N} \left(\frac{1}{2} X_k e^{j\omega_k t} + \frac{1}{2} X_k^* e^{-j\omega_k t}\right).$$

Each sinusoid is decomposed into two rotating phasors: one at the positive frequency $+\omega_k$ and one at the negative frequency $-\omega_k$. The spectrum is defined as a sequence of frequency–amplitude pairs:

$$\mathcal{S}_x = \{(\omega_k, 1/2\, X_k), (-\omega_k, 1/2\, X_k^*)\}_{k=1}^{N}.$$

The spectrum is the frequency-domain representation of the signal.

Introduce a_k as the complex amplitude used in the spectrum:

$$a_k = \frac{1}{2} X_k = \frac{A_k}{2} e^{j\phi_k}.$$

The spectrum can then be described as a set of pairs (ω_k, a_k).

3.1.1 Graphing the spectrum

Each frequency component is represented by a vertical line at its frequency, with line length proportional to the magnitude $|a_k|$.

Example. For x(t)=3 cos(2π·100 t)+2 cos(2π·250 t), the spectrum contains lines at ±100 Hz with magnitude 1.5 and at ±250 Hz with magnitude 1. This makes the frequency-domain picture much easier to read than the time waveform alone.

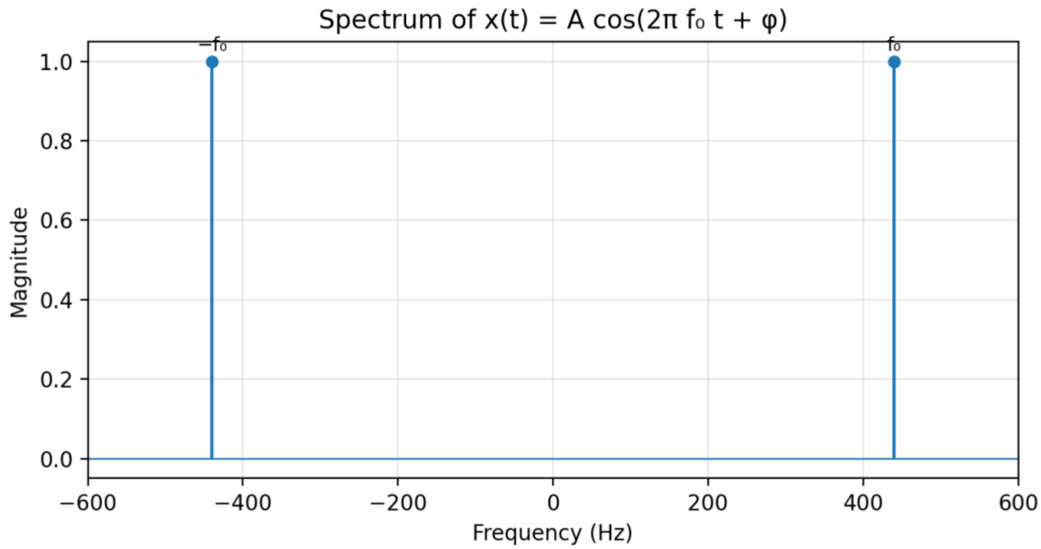

Figure 3.1. Spectrum of a sinusoidal signal.

3.2 Beat notes

Multiplying two sinusoids of different frequencies can create an interesting auditory effect called a **beat note**. Multiplication of sinusoids is also used in amplitude modulation (AM) for radio transmission.

3.2.1 Multiplication of sinusoids

Define a beat-note signal as

$$x(t) = \cos(2\pi \cdot 5t)\cos(2\pi \cdot 0.5t).$$

To define its spectrum, rewrite the product as a sum by using

$$\cos\alpha\cos\beta = \frac{1}{2}\cos(\alpha+\beta) + \frac{1}{2}\cos(\alpha-\beta).$$

Thus,

$$x(t) = \frac{1}{2}\cos(2\pi \cdot 5.5t) + \frac{1}{2}\cos(2\pi \cdot 4.5t).$$

The four spectral components occur at 5.5, 4.5, -4.5, and -5.5 Hz. Neither of the two original frequencies used in the product appears directly in the spectrum.

3.2.2 Beat-note waveform

Beat notes are commonly produced by adding two sinusoids whose frequencies are close:

$$x(t) = A\cos(2\pi(f_c + f_d)t) + A\cos(2\pi(f_c - f_d)t),$$

where f_c is the centre frequency and f_d is a much smaller deviation frequency. The spectrum contains components at $f_c + f_d$ and $f_c - f_d$, together with their negative-frequency counterparts.

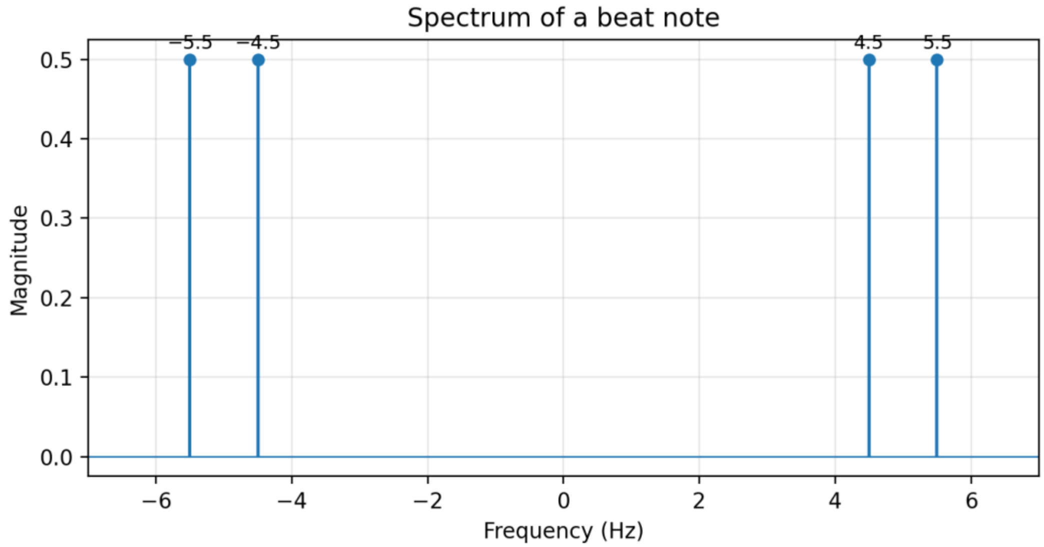

Figure 3.2. Spectrum of a beat note.

Using the sum-to-product identity,

$$x(t) = 2A\cos(2\pi f_d t)\cos(2\pi f_c t).$$

The high-frequency cosine at f_c is multiplied by a slowly varying envelope $2A\cos(2\pi f_d t)$. The perceived beat rate is $2f_d$, because the magnitude of the envelope reaches a maximum twice per envelope cycle.

Example. If x(t)=cos(2π·4.5 t)+cos(2π·5.5 t), then x(t)=2 cos(2π·0.5 t) cos(2π·5 t). The fast oscillation is centred at 5 Hz, while the envelope reaches a peak every 1 second, which is heard as the beat rate.

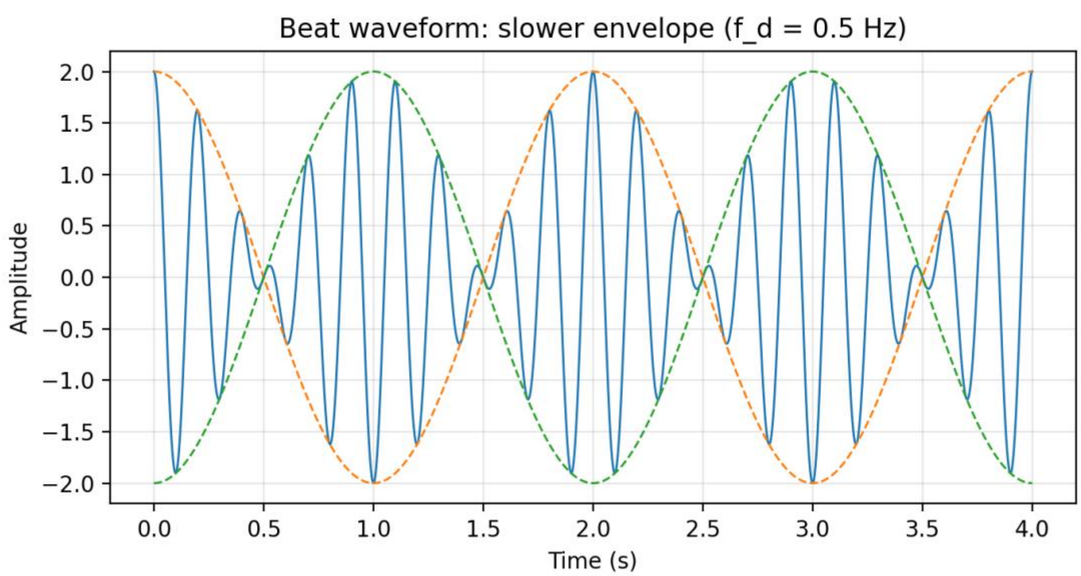

Figure 3.3. Beat-note waveform for one choice of centre and deviation frequencies.

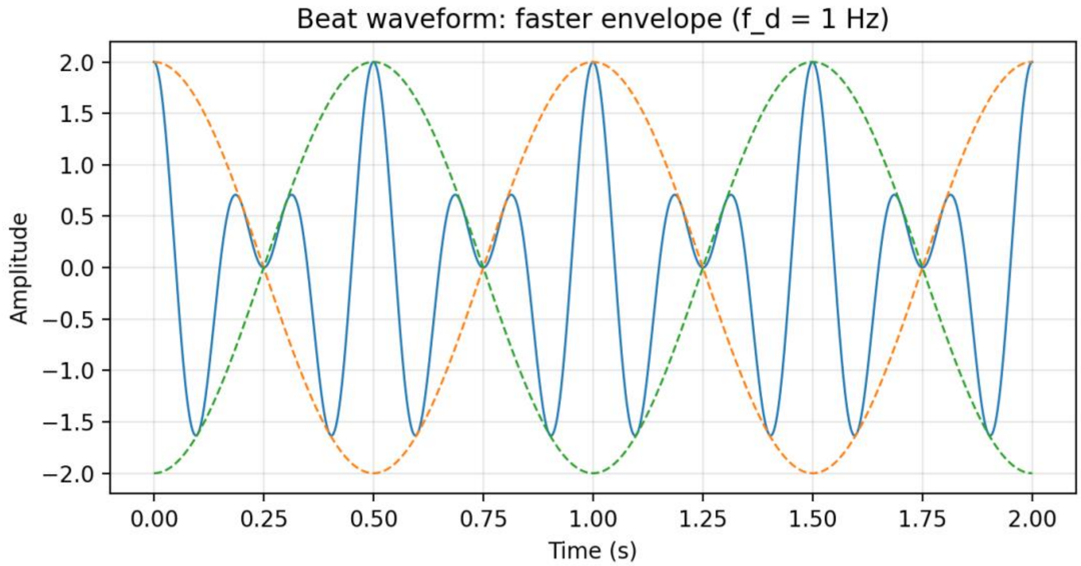

Figure 3.4. Beat-note waveform for a different deviation frequency.

3.2.3 Amplitude modulation

Amplitude modulation (AM) multiplies a low-frequency signal by a high-frequency sinusoid:

$$x(t) = [A + v(t)]\cos(2\pi f_c t),$$

where $v(t)$ is the voice or music signal and f_c is the carrier frequency. The carrier frequency is much higher than any frequency present in $v(t)$.

For example, if

$$v(t) = B\cos(2\pi f_m t),$$

then

$$x(t) = A\cos(2\pi f_c t) + \frac{B}{2}\cos(2\pi(f_c + f_m)t) + \frac{B}{2}\cos(2\pi(f_c - f_m)t).$$

The principal difference between an AM signal and a beat note is that the AM envelope does not reach zero when $A > |v(t)|$. The AM spectrum contains a strong carrier component at f_c and two sidebands at $f_c \pm f_m$.

Example. With v(t)=1+0.5 cos(2π·100 t) and carrier frequency f_c=1000 Hz, the AM signal has a carrier line at 1000 Hz and two sidebands at 900 Hz and 1100 Hz. This is the standard three-line AM spectrum shown in communication systems.

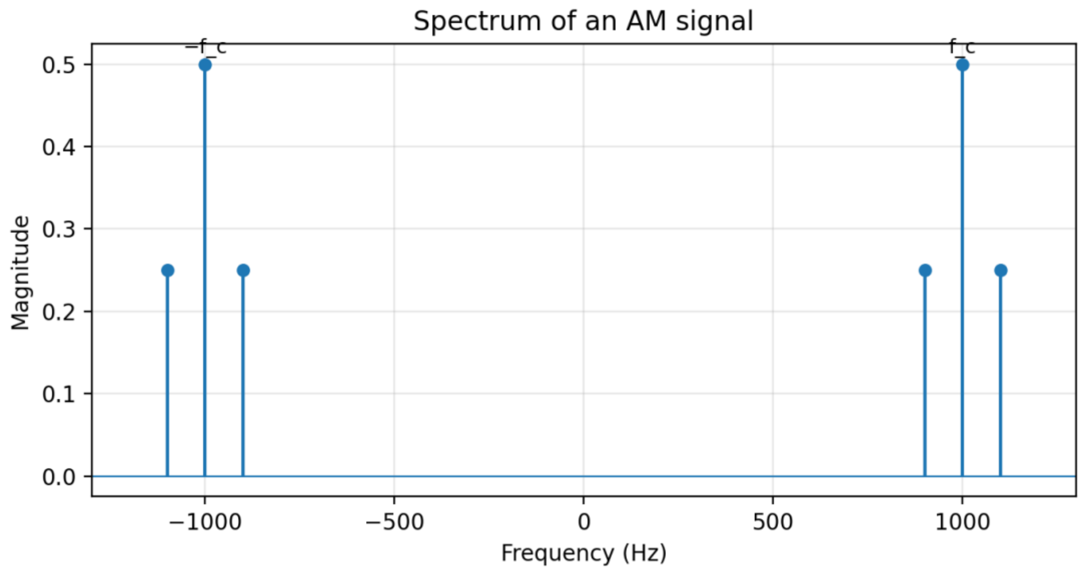

Figure 3.5. Spectrum of an AM signal.

3.3 Periodic waveforms

Two or more harmonically related cosines can be represented as

$$x(t) = \sum_{k=0}^{N} A_k \cos(k\omega_0 t + \phi_k),$$

where $k\omega_0$ are harmonic frequencies. The component at $k\omega_0$ is called the kth harmonic because it is an integer multiple of the fundamental frequency ω_0.

Using phasors,

$$x(t) = \Re\left\{\sum_{k=0}^{N} X_k\, e^{jk\omega_0 t}\right\}, \qquad X_k = A_k e^{j\phi_k}.$$

The fundamental period is

$$T_0 = \frac{2\pi}{\omega_0} = \frac{1}{f_0},$$

and $x(t + T_0) = x(t)$ for all t.

3.3.1 Synthetic vowel

A periodic signal can be synthesized as

$$x(t) = \Re\left\{\sum_{k=1}^{N} X_k\, e^{j2\pi k f_0 t}\right\},$$

where the fundamental frequency is $f_0 = 100$ Hz. Selected complex amplitudes used to approximate the vowel “a” are listed below.

k	F_k (Hz)	X_k	Magnitude	Phase (rad)
1	100	0	0	0
2	200	$771 + j12202$	12,226	1.508
3	300	0	0	0
4	400	$-8865 + j28048$	29,416	1.876

k	F_k (Hz)	X_k	Magnitude	Phase (rad)
5	500	$48001 - j8995$	48,836	-0.185
6	600	0	0	0
⋮	⋮	⋮	⋮	⋮
15	1500	0	0	0
16	1600	$1657 - j13520$	13,621	-1.449
17	1700	$4723 + j0$	4,723	0

Table 3.1. Complex amplitudes of a periodic signal that approximates the vowel "a".

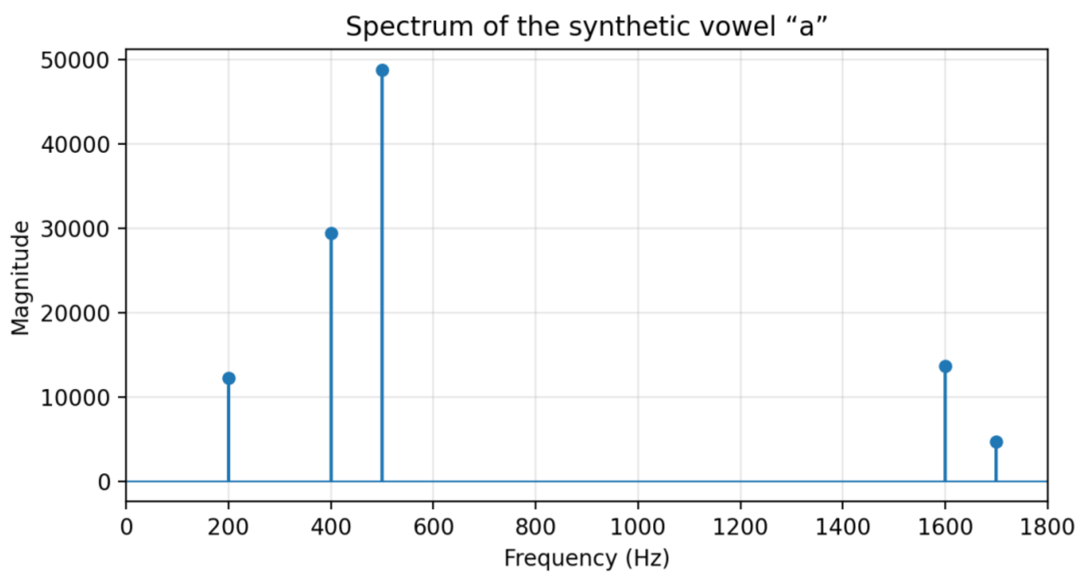

Figure 3.6. Spectrum of the signal defined by Table 3.1.

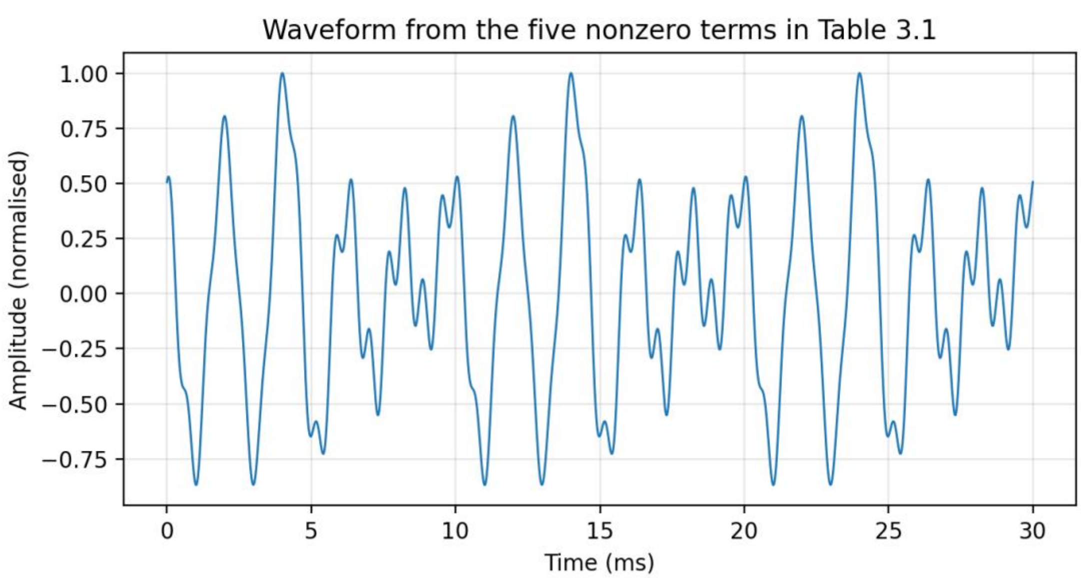

Figure 3.7. Sum of the five nonzero terms in Table 3.1.

3.3.2 Example of a nonperiodic signal

The sinusoidal synthesis formula

$$x(t) = \sum_{k=1}^{N} A_k \cos(2\pi f_k t + \phi_k)$$

remains valid even when the frequencies f_k are not harmonically related. To illustrate the relation between harmonic frequencies and periodicity, consider

$$x_1(t) = \cos(2\pi f_0 t) + \frac{1}{3}\cos(2\pi \cdot 3 f_0 t) + \frac{1}{5}\cos(2\pi \cdot 5 f_0 t).$$

A slightly perturbed signal is

$$x_2(t) = \cos(2\pi f_0 t) + \frac{1}{3}\cos(2\pi (3 f_0 + \Delta f_1) t) + \frac{1}{5}\cos(2\pi (5 f_0 + \Delta f_2) t).$$

The amplitudes are unchanged, but the frequencies are no longer exact integer multiples of one common fundamental.

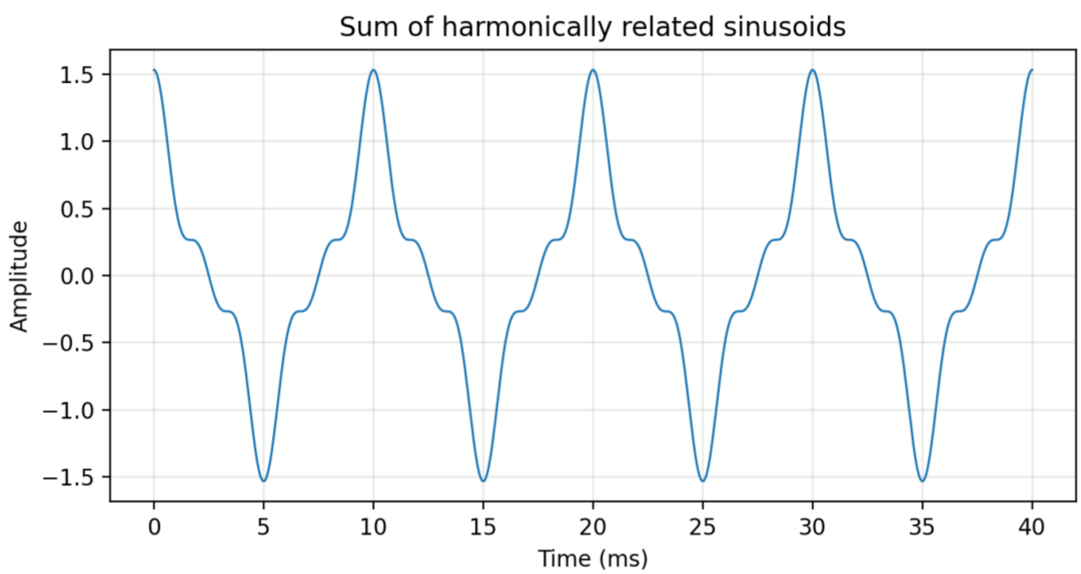

Figure 3.8. Sum of harmonically related sinusoids.

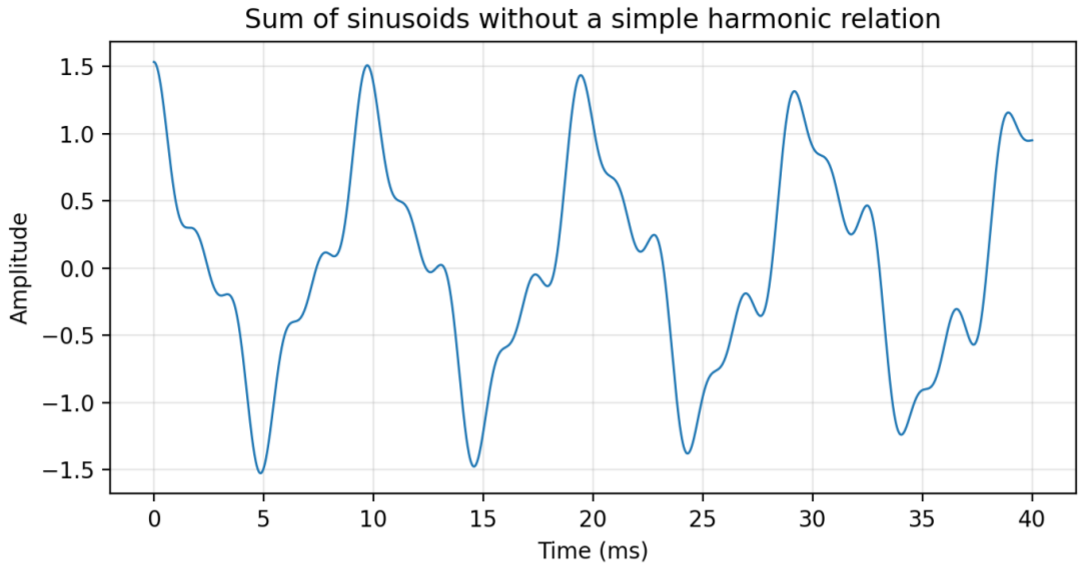

Figure 3.9. Sum of three sinusoids without a simple harmonic relationship.

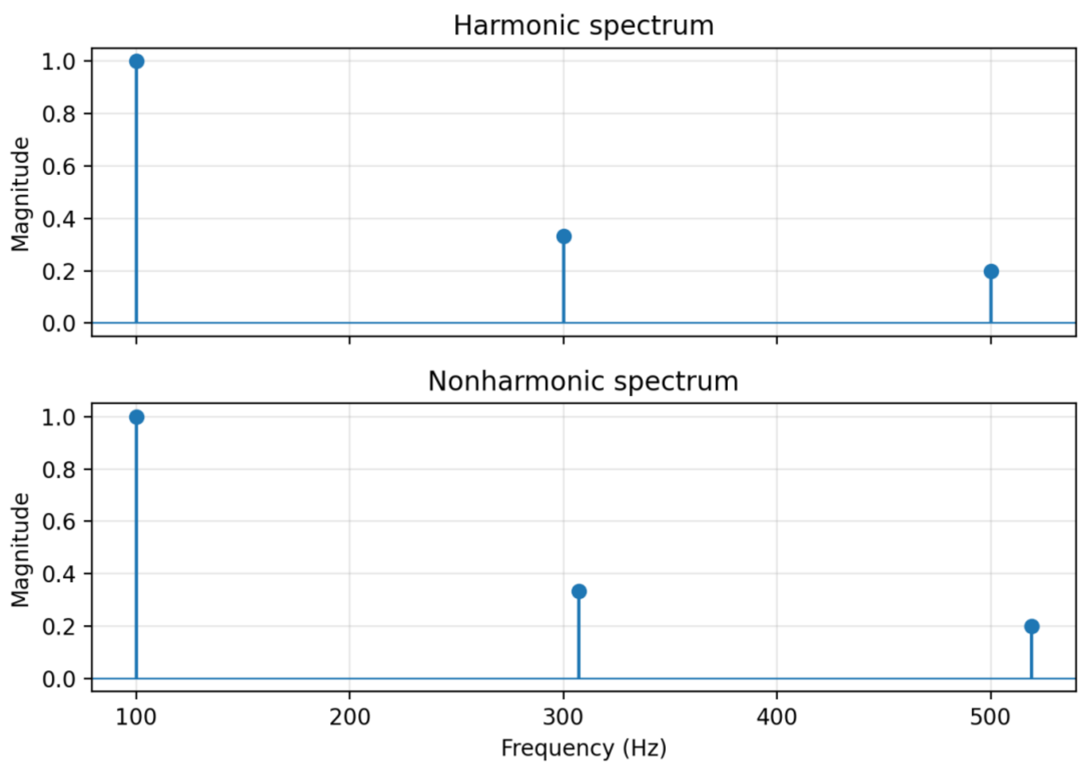

Figure 3.10. Spectra of the signals in Figures 3.8 and 3.9.

3.4 Fourier series

The examples above show that periodic waveforms can be synthesized from harmonically related sinusoids. Fourier-series theory states that a broad class of periodic signals can be represented by an infinite sum of harmonically related complex exponentials:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k\, e^{jk\omega_0 t}, \qquad \omega_0 = \frac{2\pi}{T_0}.$$

The exponential indexed by k has frequency

$$f_k = \frac{k}{T_0} = k f_0.$$

There are two complementary operations:

- **Fourier analysis:** start with $x(t)$ and calculate the coefficients a_k.
- **Fourier synthesis:** start with the coefficients a_k and generate $x(t)$.

For a real signal, the coefficients have conjugate symmetry:

$$a_{-k} = a_k^*.$$

The synthesis equation can therefore be written as

$$x(t) = a_0 + 2\sum_{k=1}^{\infty} |a_k| \cos(k\omega_0 t + \angle a_k).$$

3.4.1 Fourier-series analysis

The complex amplitudes of any periodic signal can be calculated using the Fourier integral:

$$a_k = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} x\,(t) e^{-jk\omega_0 t}\, dt.$$

The DC component is

$$a_0 = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} x\,(t)\, dt,$$

which is the average value of the signal over one period. The integral is appropriate when a formula for $x(t)$ is available. When the signal is known only through samples or a recording, numerical methods are required.

3.4.2 Derivation of the Fourier series

The derivation relies on the fact that the integral of a complex exponential over an integer number of periods is zero:

$$\int_{t_0}^{t_0+T_0} e^{jk\omega_0 t}\, dt = 0, \qquad k \neq 0.$$

Indeed,

$$\int_{t_0}^{t_0+T_0} e^{jk\omega_0 t}\, dt = \left.\frac{e^{jk\omega_0 t}}{jk\omega_0}\right|_{t_0}^{t_0+T_0} = \frac{e^{jk\omega_0 t_0}\left(e^{jk2\pi} - 1\right)}{jk\omega_0} = 0.$$

Define the complex exponential basis functions

$$\phi_k(t) = e^{jk\omega_0 t}.$$

All of them repeat with period T_0:

$$\phi_k(t + T_0) = \phi_k(t).$$

Their orthogonality property is

$$\frac{1}{T_0}\int_{t_0}^{t_0+T_0} \phi_k\,(t)\phi_\ell^*(t)\, dt = \begin{cases} 1, & k = \ell, \\ 0, & k \neq \ell. \end{cases}$$

Assuming

$$x(t) = \sum_{k=-\infty}^{\infty} a_k \, \phi_k(t),$$

multiply both sides by $\phi_\ell^*(t)$ and integrate over one period:

$$\frac{1}{T_0}\int_{t_0}^{t_0+T_0} x\,(t)\phi_\ell^*(t)\,dt = \sum_{k=-\infty}^{\infty} a_k \frac{1}{T_0}\int_{t_0}^{t_0+T_0} \phi_k\,(t)\phi_\ell^*(t)\,dt = a_\ell.$$

Hence the Fourier-analysis and Fourier-synthesis equations are

$$\boxed{a_k = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} x\,(t)e^{-jk\omega_0 t}\,dt}$$

and

$$\boxed{x(t) = \sum_{k=-\infty}^{\infty} a_k \, e^{jk\omega_0 t}}.$$

3.5 Spectrum of a Fourier series

The coefficients a_k are the complex amplitudes that define the spectrum of $x(t)$. Consider

$$x(t) = \sin^3(3\pi t).$$

Using the inverse Euler formula,

$$x(t) = \left(\frac{e^{j3\pi t} - e^{-j3\pi t}}{2j}\right)^3$$

and expanding,

$$x(t) = \frac{j}{8}e^{j9\pi t} - \frac{3j}{8}e^{j3\pi t} + \frac{3j}{8}e^{-j3\pi t} - \frac{j}{8}e^{-j9\pi t}.$$

The fundamental radian frequency is $\omega_0 = 3\pi$, so the Fourier coefficients are

$$a_k = \begin{cases} 0, & k = 0, \\ -\frac{3j}{8}, & k = 1, \\ \frac{3j}{8}, & k = -1, \\ \frac{j}{8}, & k = 3, \\ -\frac{j}{8}, & k = -3, \\ 0, & \text{otherwise.} \end{cases}$$

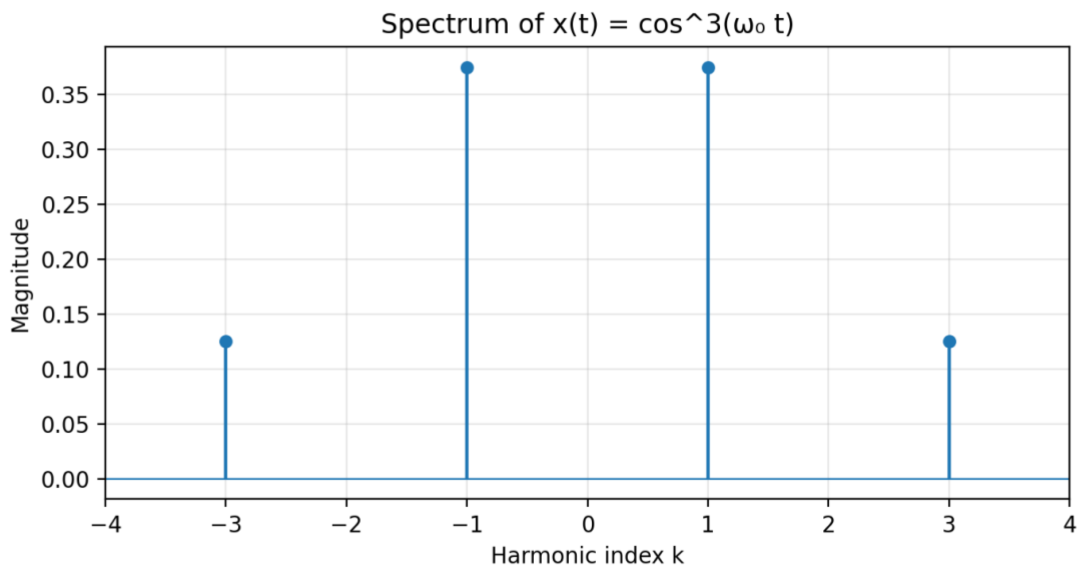

Figure 3.11. Spectrum of a cubed sinusoid derived from its Fourier-series coefficients.

3.6 Fourier analysis of periodic signals

3.6.1 The rectangular pulse train

Consider one period of a rectangular waveform:

$$s(t) = \begin{cases} 1, & 0 \le t < T_1, \\ 0, & T_1 \le t < T_0, \end{cases} \qquad s(t + T_0) = s(t).$$

Substituting into the Fourier integral gives

$$a_k = \frac{1}{T_0} \int_0^{T_1} e^{-jk\omega_0 t}\, dt.$$

For $k \neq 0$,

$$a_k = \frac{1 - e^{-jk\omega_0 T_1}}{jk\omega_0 T_0} = \frac{T_1}{T_0} \operatorname{sinc}\left(k \frac{T_1}{T_0}\right) e^{-jk\pi T_1/T_0},$$

where

$$\operatorname{sinc}(u) = \frac{\sin(\pi u)}{\pi u}.$$

The DC coefficient must be evaluated separately:

$$a_0 = \frac{1}{T_0} \int_0^{T_1} 1\, dt = \frac{T_1}{T_0}.$$

Thus,

$$a_k = \begin{cases} \dfrac{T_1}{T_0}, & k = 0, \\ \dfrac{T_1}{T_0} \operatorname{sinc}\left(k \dfrac{T_1}{T_0}\right) e^{-jk\pi T_1/T_0}, & k \neq 0. \end{cases}$$

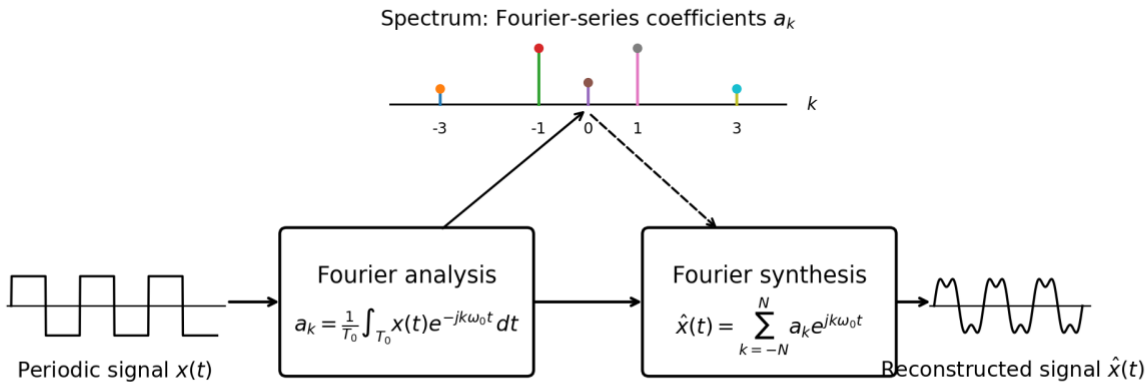

Figure 3.12. Main elements of Fourier analysis and synthesis.

Other useful examples include the triangular wave and nonperiodic signals.

Example. For a 50% duty-cycle rectangular wave, the Fourier-series coefficients decay approximately as 1/|k| and only the odd harmonics are present. This explains the sharp edges of the waveform: many harmonics are needed to reconstruct them accurately.

3.5 Time-frequency spectrum

So far, amplitudes, frequencies, and phases have been assumed constant. Most natural sounds, however, change substantially over time. This leads to the concept of a **time-frequency spectrum**, or **spectrogram**.

A familiar way to develop intuition about a spectrogram is musical notation: horizontal position represents time, vertical position represents pitch or frequency, and the written symbols indicate the presence and duration of sound events.

Figure 3.13. Example of musical notation as an intuitive time-frequency representation.

3.6 Frequency modulation: chirp signals

3.6.1 A signal with linearly changing frequency

A **chirp** is a signal whose frequency changes continuously from a low value to a high value. Concatenating many short constant-frequency sinusoids would create discontinuities. A better method allows the sinusoidal phase to vary with time:

$$x(t) = A\cos\big(\theta(t)\big),$$

where $\theta(t)$ is the time-varying phase. For a quadratic phase,

$$\theta(t) = \omega_0 t + \frac{1}{2}\alpha t^2 + \phi_0.$$

The instantaneous radian frequency is the slope of the phase:

$$\omega_i(t) = \frac{d\theta(t)}{dt}.$$

In hertz,

$$f_i(t) = \frac{1}{2\pi}\frac{d\theta(t)}{dt}.$$

For the quadratic phase above,

$$\omega_i(t) = \omega_0 + \alpha t, \qquad f_i(t) = f_0 + \frac{\alpha}{2\pi}t.$$

Thus, a quadratic phase produces a frequency that varies linearly with time. Frequency variation produced through a changing phase is called **frequency modulation**.

Example. A linear chirp can be built so that the instantaneous frequency rises from 220 Hz to 2320 Hz over 2 seconds. In the spectrogram, this produces an upward diagonal ridge, which directly visualises the frequency modulation.

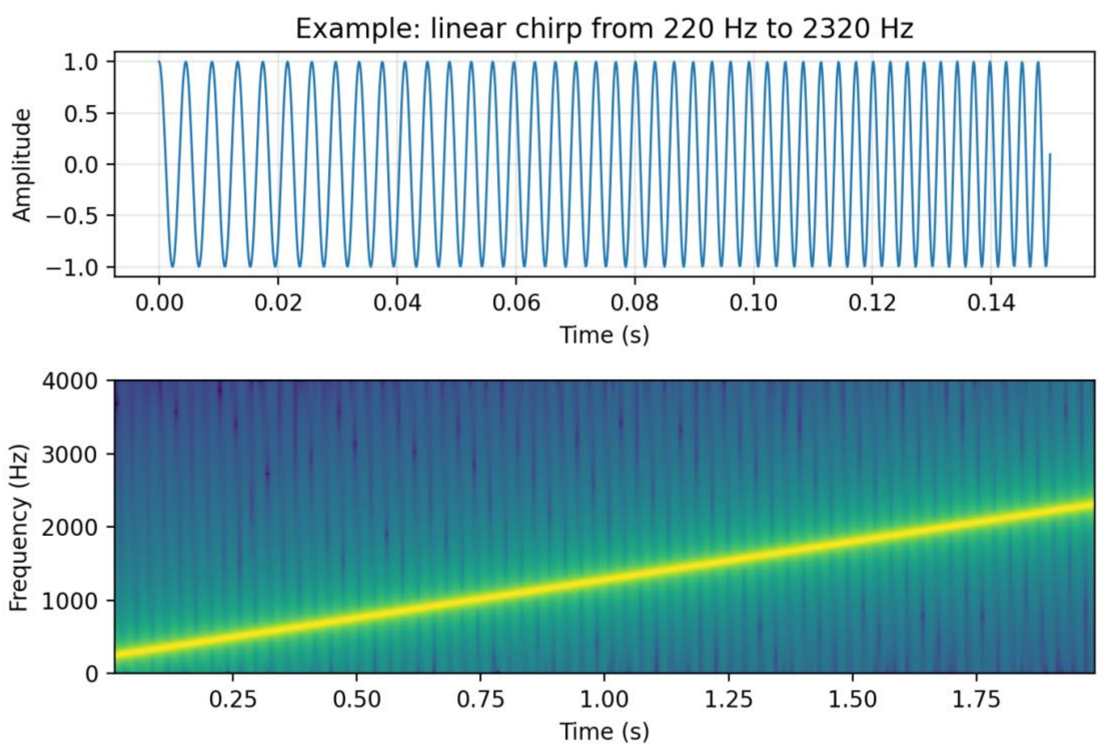

Figure 3.14. Linear chirp signal and its spectrogram.

3.7 Notation

Symbol	Meaning
$x(t)$	Continuous-time signal
X_k	Complex amplitude (phasor) of the kth sinusoidal component
ω_k	Radian frequency of the kth component (rad/s)
f_k	Cyclic frequency of the kth component (Hz)
φ_k	Initial phase of the kth component (rad)
ω_0, f_0	Fundamental radian and cyclic frequencies
T_0	Fundamental period, $T_0 = 1/f_0$
a_k	Complex Fourier-series coefficient
f_c	Carrier frequency in amplitude modulation
f_d	Deviation frequency used in the beat-note representation
$\Phi(t)$	Time-varying phase of a frequency-modulated signal
$\omega_i(t)$	Instantaneous radian frequency, $d\Phi(t)/dt$